American International University-Bangladesh (AIUB)
Faculty of Engineering
Department of CSE, EEE, and CoE

EEE4103 MICROPROCESSOR AND EMBEDDED SYSTEM
COURSE CAPSTONE PROJECT PROPOSAL FORM

SEMESTER: FALL 2024-25
PROJECT TITLE: (must be an embedded system designed using Arduino/STM32/Raspberry Pi with other necessary sensors, actuators, components, etc. Both hardware implementation and simulation must be shown by the end of the semester. However, the proposal form should be submitted within the next two weeks from now on) 2 MARKS
SURVEY: (to develop a process for complex engineering problems considering cultural and societal factors (use Google or Microsoft Forms to design questions, show the outcomes in the form of a pie/bar chart, show the links of questions, minimum 10 questions, and 40 respondents) 5 MARKS
AIMS AND OBJECTIVES OF THE PROJECT: (there must be a minimum of three aims and objectives of your work using measurable action verbs of Bloom's Taxonomy, for example, design, simulate, implement, analyze, etc.) 3 MARKS
LITERATURE REVIEW: (must be related to embedded system design with other necessary citations from the most recently published papers. At least 5 most recent publications must be collected, studied, identified the problems, and then discussed in this section) 5 MARKS
EXPERIMENTAL BLOCK DIAGRAM: (demonstrate how this project is inter-connected with various components and how various signals flow among them using text boxes and arrows) 4 MARKS
POSSIBLE OUTCOMES OF THE PROJECT: (describe how this project can help society and improve the culture of the society) 4 MARKS
PROJECT TIMELINE (GANTT CHART): describe how this project will progress with time by breaking the whole task into several components concerning some specific time using a horizontal or a vertical bar graph) 5 MARKS
REFERENCES: (only published paper-based references are allowed with the IEEE format, don't use YouTube links, Wikipedia, or any random websites for references, however, datasheets, databases, can be used as references with the appropriate formats, links, and access dates): 2 MARKS **Sample for the conference paper:** [1] M. H. Bhuyan and Q. D. M. Khosru, "Linear Asymmetric Pocket Profile Based Pinch Off Voltage Model for Nano Scale n-MOSFET," Proceedings of the IEEE sponsored International Conference on Electrical, Computer and Communication Engineering (ICECCE2017), organized by the Chittagong University of Engineering and Technology (CUET), Cox's Bazar, Bangladesh, 16-18 February 2017, pp. 28-32. **Sample for the journal paper:** [1] M. H. Bhuyan and Q. D. M. Khosru, "Effects of Temperature on Reverse Short Channel Effect in Pocket Implanted Sub-100 nm n-MOSFET," Journal of Materials Science and Engineering, USA, 1934-8959, vol. 4, no, 7, July 2010, pp. 18-23, DOI:10.17265/2161-6213/2010.07.004.

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Instructions:

1. There is no definite format to write the proposal, but students must follow the mentioned instructions properly.
2. Fill in the form accurately with all necessary information.
3. Make a color print on this form.
4. Figures, tables, charts, circuit diagrams, block diagrams, and wave shapes must be color printed.
5. The survey form link must be provided in the proposal form with the answers.

FOR FACULTY USE ONLY

COMMENTS BY COURSE TEACHER:

Prof. Dr. Engr. Muhibul Haque Bhuyan

COURSE TEACHER'S NAME **COURSE TEACHER'S SIGNATURE** **DATE**

GROUP MEMBERS

(Maximum 6 students are permitted to carry out a single Project. However, depending on the capability of the students, 4 students may be allowed but not less than that)

NAME: SASHYNI ISLAM **ID:** 21-4491-2 **PROGRAM:** EEE **EMAIL:** 21-4491-2@student.aiub.edu	**NAME**: MD. EMRAN NAZIR EFITY **ID:** 22-47802-2 **PROGRAM:** CSE **EMAIL:** 22-47802-2@student.aiub.edu
NAME: RAYSUL AZAD TURZO **ID #:** 22-47820-2 **PROGRAM:** CSE **EMAIL:** 22-47820-2@student.aiub.edu	**NAME**: MD. TANVIRUL ISLM **ID #:** 22-48000-2 **PROGRAM:** CSE **EMAIL:**22-48000-2@student.aiub.edu
NAME: NAZIFA TABASSUM TANDRA **ID #:** 22-49977-3 **PROGRAM:** CSE **EMAIL:** 22-48006-2@aiub.student.edu	**NAME**: …… **ID #:**…… **PROGRAM:** EEE/CoE/CSE **EMAIL:**
REMARKS (for OFFICE use only)	

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Course Name:	Microprocessor and Embedded System	**Course Code:**	EEE 4103
Semester:	Fall 2024-2025	**Sec:**	D
Faculty Member:	PROTIK PARVEZ SHEIKH		

Capstone Project Title:	IoT based Air Quality and Weather Monitoring System for Sustainable Urban Environment
Project Group No.	01

Sl	Student ID #	Student Name	Obtained Marks
1.	**21-44901-2**	SASHYNI ISLAM	
2.	**22-47802-2**	MD. EMRAN NAZIR EFITY	
3.	**22-47820-2**	RAYSUL AZAD TURZO	
4.	**22-48000-2**	MD. TANVIRUL ISLAM	
5.	**22-48006-2**	NAZIFA TABASSUM TADRA	

Assessment Materials and Marks Allocation:

COs	Assessment Materials	POIs	Marks
CO3	Course Capstone Proposal Form	P.c.2.C6	30

Assessment Rubrics:

KPIs	Excellent [2]	Proficient [1.5]	Good [1]	Acceptable [0.5]	Unacceptable [0]	No Response [0]	Secured Marks
Project Title	The title reflects an issue related to complex engineering problems showing targets and methods with possible outcomes.	The title reflects an issue related to complex engineering problems showing targets and methods but some missing issues.	The title reflects an issue related to the course capstone project but there may be some missing issues.	The title reflects an issue related to the course capstone project but is not complete or specific.	The title does not reflect any issues related to the course capstone project.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (2)**	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Survey	The survey developed as a process for complex engineering problems considering cultural and societal factors has superior variables, targets, measures, and the implementation process is clear and challenging for future project implementation with several possible outcomes having good impacts.	The survey developed as a process for complex engineering problems considering cultural and societal factors has good variables, targets, measures, and the implementation process is clear and challenging for future project implementation with some possible outcomes with little impact.	The survey developed as a process for complex engineering problems considering cultural and societal factors has moderate variables, targets, measures, and the implementation process is clear and challenging for future project implementation with a few possible outcomes with impacts.	The survey developed as a process for complex engineering problems considering cultural and societal factors has good variables, targets, measures, and the implementation process is somewhat clear for future project implementation with very few possible outcomes with little impact.	The survey developed as a process for complex engineering problems considering cultural and societal factors has poor variables, targets, measures, and the implementation process is very unclear for future project implementation with a few possible outcomes but no impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (5)**	

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KPIs	Excellent [3]	Proficient [2.5]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Aims and Objectives	Aims and objectives are written to solve complex engineering problems considering cultural and societal factors with specific targets, measurement, and implementation processes that are clear and challenging and have several possible outcomes having very good impacts.	Aims and objectives are written to solve complex engineering problems considering cultural and societal factors with general targets, measurement, and implementation processes that are not clear and challenging and have some possible outcomes having good impacts.	Aims and objectives are written to solve complex engineering problems considering a few cultural and societal factors with narrow targets; measurement, and implementation processes are clear and challenging and have a few possible outcomes having some impacts.	Aims and objectives are written to solve complex engineering problems considering cultural or societal factors with a very target; measurement and implementation processes are not clear or challenging and have little possible outcome having no impact.	Aims and objectives are written to solve complex engineering problems but do not consider cultural and societal factors with any targets; measurement, and implementation processes are not clear and challenging and no possible outcomes have no impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (3)**	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Literature Review	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly.	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly, but all issues were not addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified and analyzed the problem correctly, but all issues were not addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified but could not analyze all the problems correctly, and all issues were not addressed with relevant or intended work.	No specific formats are maintained to review and cite the literature with recent publications. Could not identify and analyze all the problems correctly, and all issues are not addressed with relevant or intended work at all.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						**Total Marks (5)**	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Experimental Block Diagram	The block diagram is drawn to show the connections of all the possible components or sub-systems to show their interdependence with all possible flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of all of the possible components or sub-systems to show their interdependence with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of most of the possible components or sub-systems to show their interdependence with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of a few possible components or sub-systems to show their interdependence with some missing flow of signals from inputs to outputs.	The block diagram is not drawn to show the connections of all possible components or sub-systems to show their interdependence and flow of signals from inputs to outputs.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (4)**	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Possible Outcomes	Outcomes are written to achieve complex engineering problems' solutions considering cultural and societal factors and showing measurement, and implementation processes to attain the outcomes with all possible impacts.	Outcomes are written to achieve complex engineering problems' solutions considering cultural and societal factors and showing measurement, and implementation processes to attain the outcomes with some impacts.	Outcomes are written to achieve complex engineering problems' solutions considering cultural and societal factors and do not show measurement, and implementation processes to attain the outcomes without showing any impacts.	Outcomes are written to achieve complex engineering problems' solutions but do not consider cultural and societal factors and do not show measurement, and implementation processes to attain the outcomes without showing any impacts.	Outcomes are not written to achieve complex engineering problems' solutions do not consider cultural and societal factors and do not show measurement, and implementation processes to attain the outcomes without showing any impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (4)**	

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KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Gantt Chart	Specific formats are maintained to draw the Gantt chart and there is the order of workflow with all work to be done.	Specific formats are maintained to draw the Gantt chart and there is the order of workflow with a few works missing.	Specific formats are maintained to draw the Gantt chart and there is the order of workflow with some works missing.	No specific formats are maintained to draw the Gantt chart and there is little order of workflow with some works missing.	No specific formats are maintained to draw the Gantt chart and there is no order of workflow with the most important works missing.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						**Total Marks (5)**	

KPIs	Excellent [2]	Proficient [1.5]	Good [1]	Acceptable [0.5]	Unacceptable [0]	No Response [0]	Secured Marks
References	Specific formats are maintained to write the references, and all are recently published journal and conference papers having no missing information.	Specific formats are maintained to write the references, and all are journal and conference papers, but some old papers have missing information.	No specific formats are maintained to write the references, and many are internet sources with several missing information and very old references.	No specific formats are maintained to write the references and most of them are internet sources with missing information.	No specific formats are maintained to write the references, and all are internet sources with missing information.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						**Total Marks (2)**	

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PROJECT TITLE: IoT based Air Quality and Weather Monitoring System for Sustainable Urban Environment

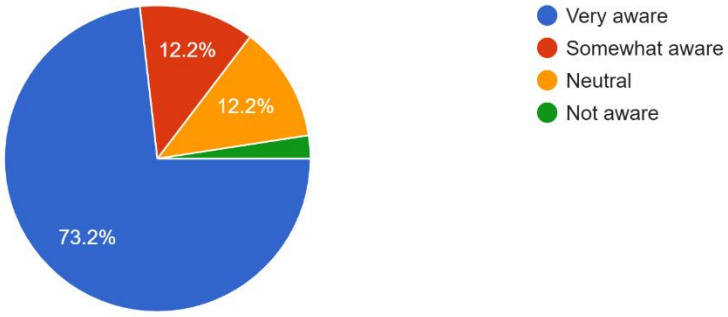
Survey:

This survey aims to gather feedback on the effectiveness and usability of an IoT-based environmental monitoring system. The system monitors air quality, temperature, humidity, and rainfall in real-time, providing data through an online dashboard and an on-site display. Warning alerts are triggered during critical air quality levels to ensure timely action. Your responses will help improve the system and assess its potential impact on raising awareness about air pollution and environmental health.

Survey link: https://docs.google.com/forms/d/1elHQ-TP4JaCtGFG6qnGA_hTc_utnaM2Um3zSDbSQTn0/edit

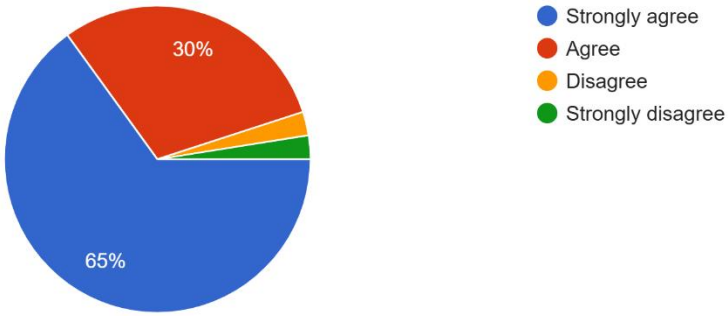
1. How aware are you of the impact of air pollution on your daily life and health?

41 responses



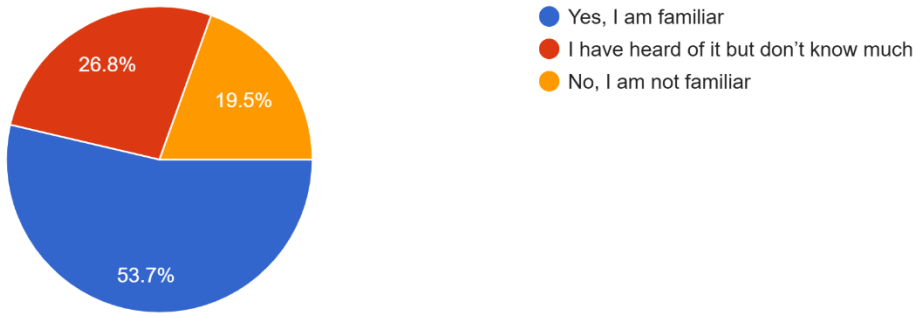
2. Do you believe monitoring air quality is essential for improving environmental conditions?

40 responses



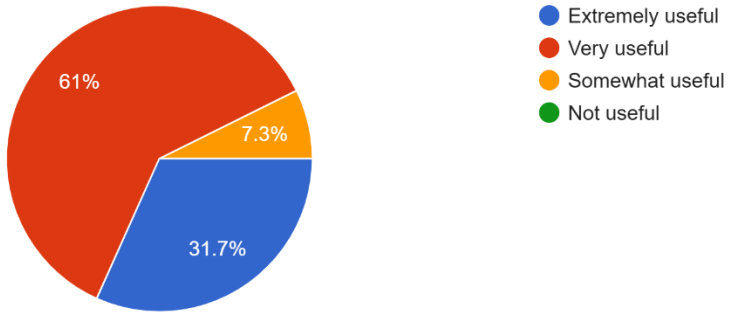
3. Are you familiar with IoT-based systems for environmental monitoring?

41 responses



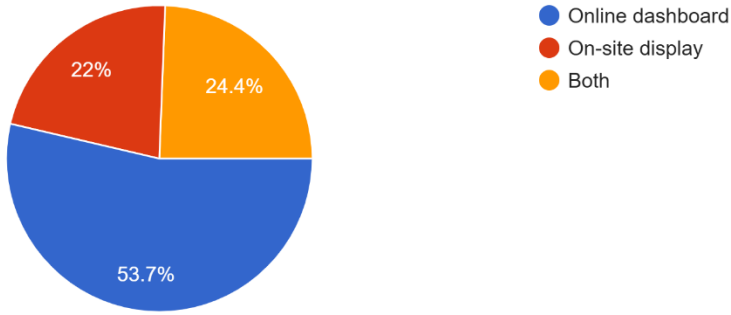
4. How useful do you think a real-time monitoring system for air quality, temperature, and humidity would be for your daily activities?

41 responses



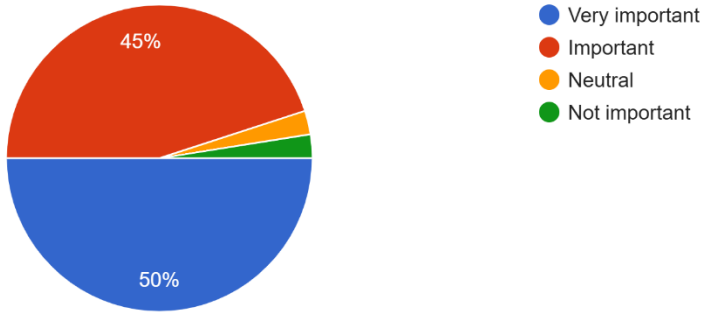
5. Would you prefer accessing environmental data through an online dashboard, an on-site display, or both?

41 responses



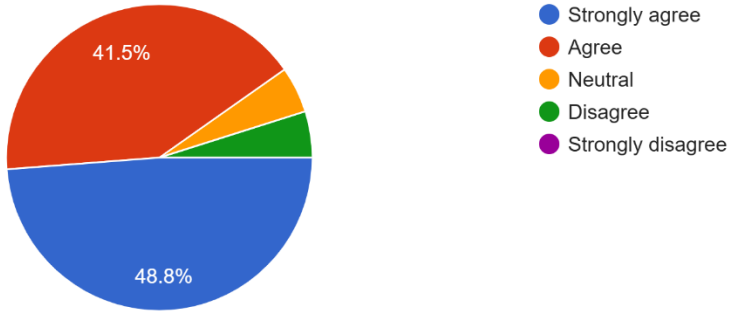
6. How important is it for you to receive warning alerts (e.g., buzzer or notifications) during critical air quality levels?

40 responses



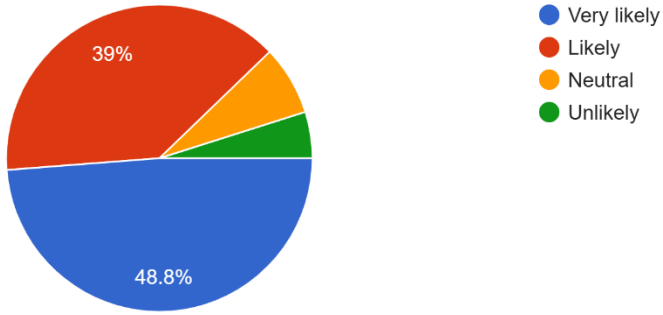
7. Do you think integrating such a monitoring system in residential or industrial areas would improve decision-making regarding environmental issues?

41 responses



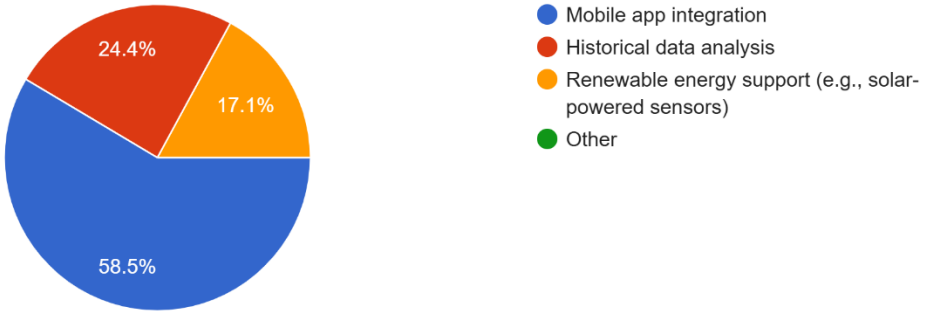
8. How likely are you to use or recommend a system like this for personal or community use?

41 responses



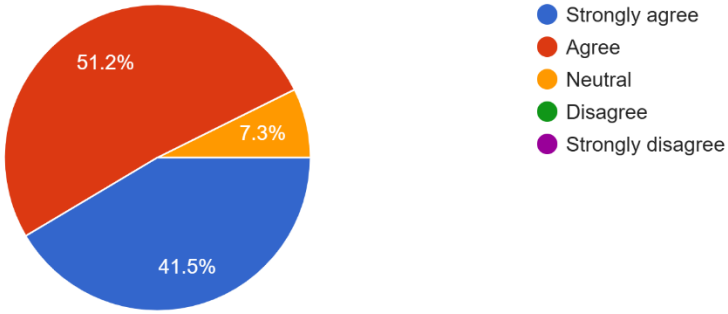
9. What additional features would you like to see in an air quality monitoring system?

41 responses



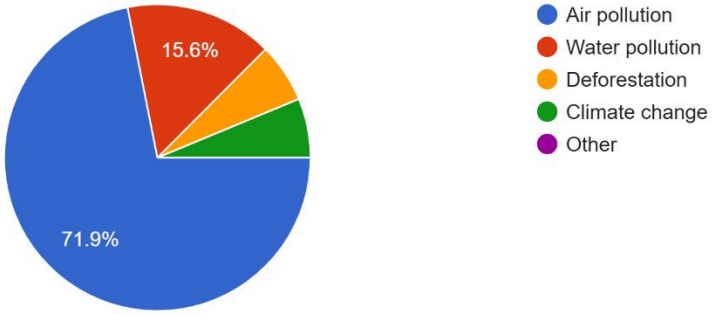
10. Do you think this system could effectively raise awareness and drive action toward combating air pollution?

41 responses



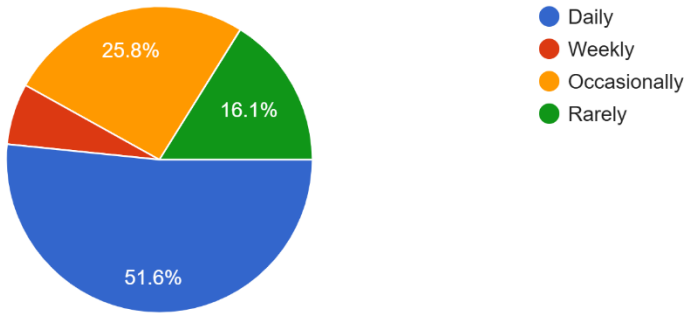
11. What do you consider the most critical environmental issue in your area?

32 responses



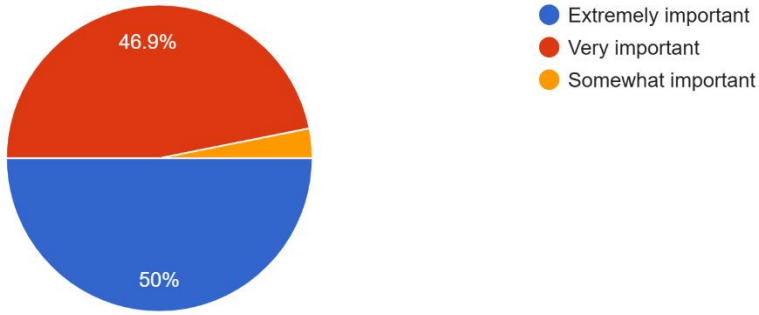
12. How often do you check or care about environmental conditions such as air quality or weather?

31 responses



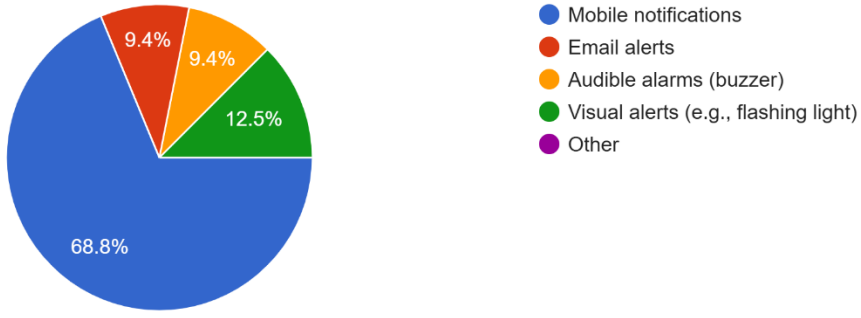
13. How important is the accuracy of environmental data for you?

32 responses



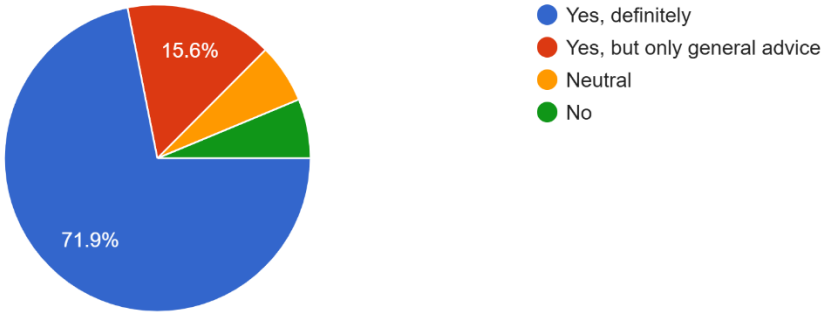
14. What type of warning alerts would you find most helpful in case of poor air quality?

32 responses



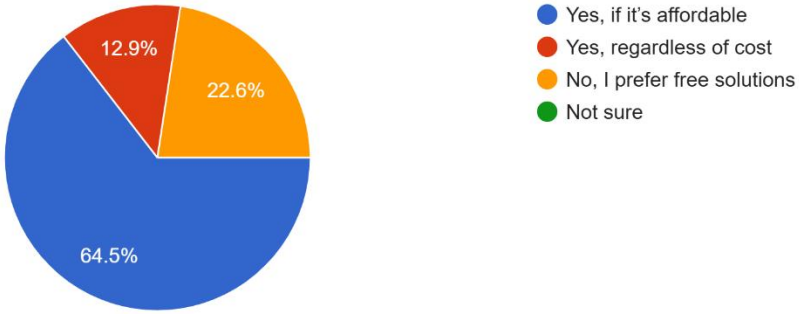
15. Would you like the system to suggest actions based on environmental conditions (e.g., reducing outdoor activities)?

32 responses



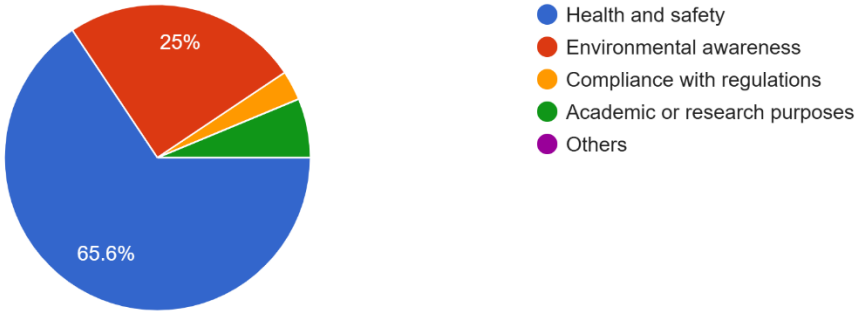
16. Would you be willing to pay for a device or service like this?

31 responses



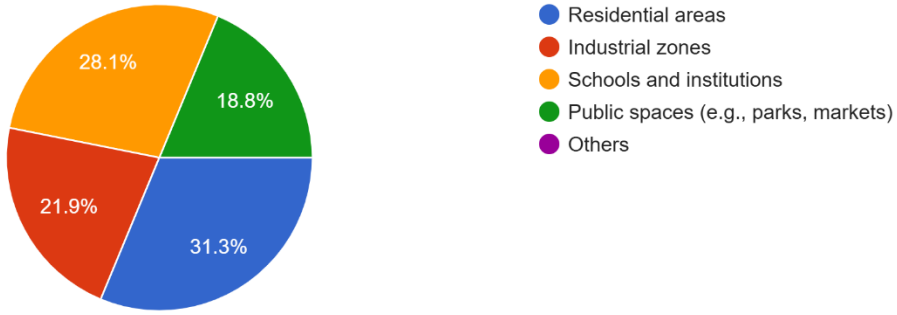
17. What would be your primary motivation for using this system?

32 responses



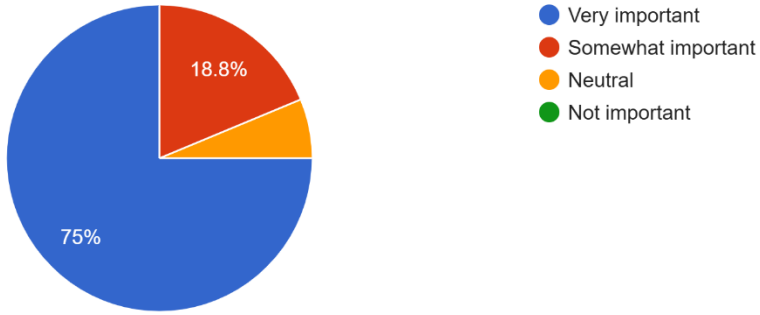
18. Where do you think such systems should be deployed primarily?

32 responses



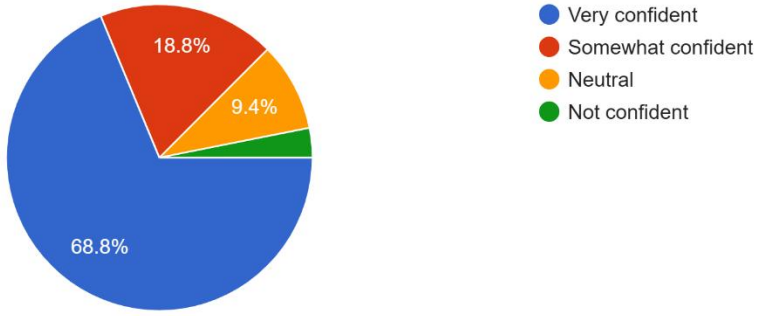
19. Do you think integrating renewable energy (e.g., solar panels) into the system is important?

32 responses



20. How confident are you that a system like this could drive change in addressing air pollution?

32 responses



AIMS AND OBJECTIVES OF THE PROJECT:

- Design and implement an Internet of Things (IoT) based system focused on monitoring air quality and weather conditions.
- Utilize specific sensors including the MQ135 70 quality sensor for detecting various air pollutants, the DHT11 sensor for measuring temperature and humidity, and a rain sensor for detecting. precipitation.
- Integrate an I2C display to provide users with real-time visual feedback on the collected data, ensuring easy interpretation and understanding.
- Incorporate a buzzer into the system to provide audible alerts in the event of gas or smoke detection, enhancing safety measures.
- Employ a NodeMCU ESP8266 Wi-Fi module to enable wireless connectivity, facilitating remote access to the system and its data.
- Develop algorithms and mechanisms for monitoring and analyzing air quality data collected by the sensors, allowing for insights into pollution levels and weather conditions.
- Implement functionality to log collected data into a cloud server, specifically Adafruit.com, leveraging an internet connection to enable storage, analysis, and sharing of the data with stakeholders and researchers

LITERATURE REVIEW: Air is crucial for life, containing vital elements necessary for health and well-being. However, rapid industrialization and the widespread use of mechanical devices have significantly increased air pollution, posing risks to health and the environment. To combat this, various air quality monitoring systems have been developed using advanced technologies and IoT integration.

Yasue Sung et al. (2019) introduced a smart indoor air quality monitoring system that measures PM10, PM2.5, CO_2, VOCs, temperature, and humidity. This system features a smartphone app for remote data access and provides daily, monthly, and yearly air quality reports. Anand Jayakumar et al. (2021) developed a three-phase air pollution monitoring system using MQ2 and MQ7 sensors with Raspberry Pi, enabling faster data processing. The system displays pollutant levels on an LCD and logs data on a web platform for remote access.

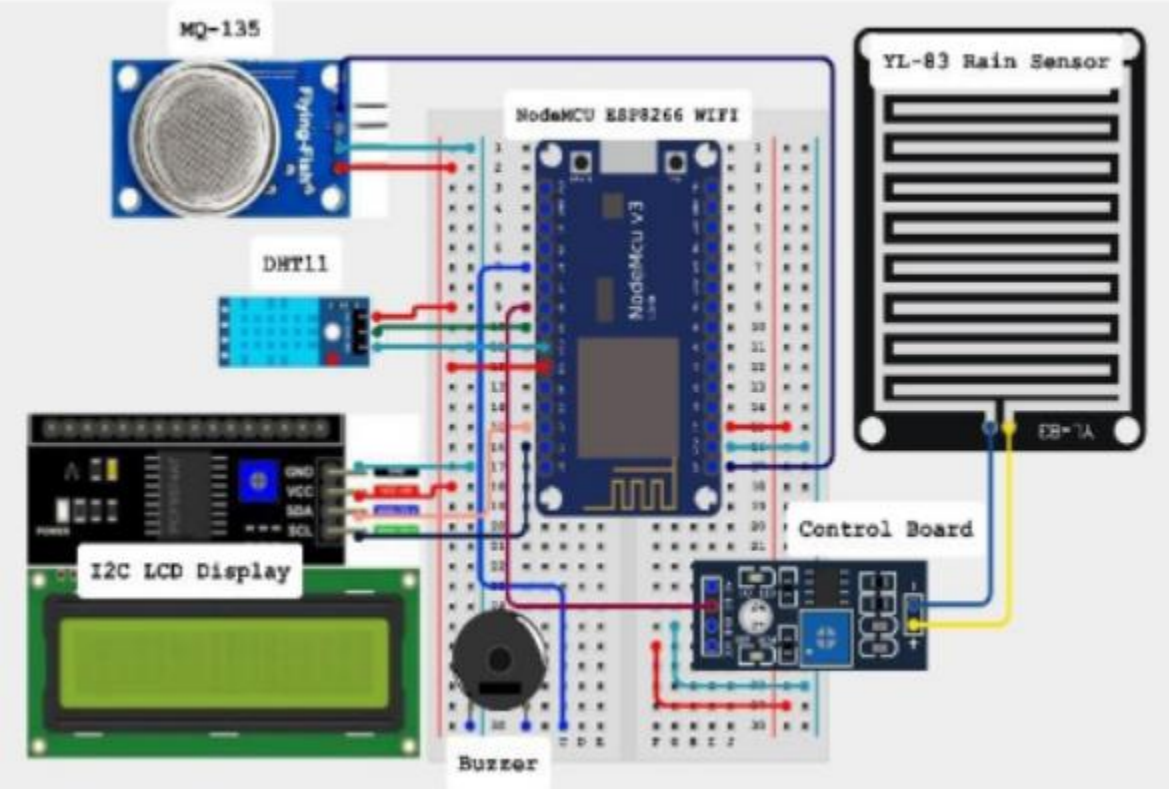
Kalangala et al. (2022) created an air quality monitoring system using the MQ135 sensor, known for its high accuracy in detecting harmful gases like CO_2, NH_3, and benzene. This system includes an alarm for critical pollution levels and offers real-time data tracking via a web server. Dinesh Kumar et al. (2021) further enhanced air quality monitoring by developing a compact, portable IoT system using Arduino and sensors to detect CO, O_3, and SO_2. The device is lightweight, cost-effective, and capable of sending alerts when pollution levels exceed safety thresholds.

These innovations demonstrate the potential of IoT-based air quality monitoring systems to provide accurate, real-time insights, raise awareness, and enable proactive measures to improve air quality and protect public health.

POSSIBLE OUTCOMES OF THE PROJECT:

1. A comprehensive analysis of existing IoT-based systems will reveal their strengths, limitations, and applications in air quality monitoring.
2. The research will identify gaps in current designs, leading to improvements in sensor accuracy, portability, and functionality.
3. It will raise awareness about air pollution and its impacts, encouraging informed decisions among individuals and policymakers.
4. A real-time monitoring system with alerts will enable immediate action during critical pollution levels.
5. The findings can be scaled for applications in homes, industries, public spaces, and portable devices.
6. The collected data will guide interventions for pollution management and influence urban planning.
7. The study will advance IoT and environmental research, fostering innovation in cost-effective and efficient solutions.

EXPERIMENTAL CIRCUIT DIAGRAM:



PROJECT TIMELINE (GNATT CHART):

	23 Nov to 28 Nov	29 Nov to 11 Dec	12 Dec to 17 Dec	17 Dec to 25 Dec	26 Dec to 4 Jan	5 Jan to 17 Jan	18 Jan to 30 Jan	31 Jan
Selection of topic & proposal								
Software Simulation and theoretical analysis								
Collection of equipment								
modification and research								
Implementation of hardware								
Verification output result modification								
Final Defense								

REFERENCES:

1. S. Singh, A. Kumar, and R. Gupta, "RFID-Based Automatic Railway Gate Control System," in 2018 3rd International Conference on Internet of Things: Smart Innovation and Usages (IoT SIU), 2018, pp. 1-5.
2. A. Sharma, S. Verma, and P. Singh, "Automatic Railway Gate Control System Using Ultrasonic Sensor," in 2019 2nd International Conference on Inventive Research in Computing Applications (ICIRCA), 2019, pp. 1-5
3. A. Kumar, S. Jain, and R. Sharma, "Vibration Analysis-Based Fault Detection System for Railway Tracks," in 2020 3rd International Conference on Communication and Electronics Systems (ICCES), 2020, pp. 1-5.
4. R. Gupta, S. Tiwari, and S. Mishra, "Vision-Based Railway Track Fault Detection System Using Image Processing Techniques," in 2017 2nd International Conference on Signal Processing and Integrated Networks (SPIN), 2017, pp. 693-698.
5. Y. Jiang, X. Wang, and Z. Li, "Integrated Automatic Railway Gate Control and Track Fault Detection System," in 2021 IEEE International Conference on Robotics and Automation (ICRA), 2021, pp. 1-6